

## SOUTH ROSEMARY, NC, USA

WMO: 724008

Lat: 36.439N

Lon: 77.710W

Elev: 78

StdP: 100.39

Time Zone: -5.00 (NAE)

Period: 99-09

WBAN: 93781

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |            |             |              |            |             | Coldest Month WS/MCDB |            |            |            | MCWS/PCWD to 99.6% DB |          | WSF          |
|---------------|-------------|-------------|-------------------------------|------------|-------------|--------------|------------|-------------|-----------------------|------------|------------|------------|-----------------------|----------|--------------|
|               |             |             | 99.6%                         |            |             | 99%          |            |             | 0.4%                  |            | 1%         |            |                       |          |              |
|               | 99.6%       | 99%         | DP                            | HR         | MCDB        | DP           | HR         | MCDB        | WS                    | MCDB       | WS         | MCDB       | MCWS                  | PCWD     |              |
| (a)<br>1      | (b)<br>-6.4 | (c)<br>-4.0 | (d)<br>-17.1                  | (e)<br>0.8 | (f)<br>-3.0 | (g)<br>-14.6 | (h)<br>1.1 | (i)<br>-1.2 | (j)<br>9.1            | (k)<br>4.7 | (l)<br>8.2 | (m)<br>5.3 | (n)<br>2.2            | (o)<br>0 | (p)<br>0.381 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 9.9                    | 34.4            | 24.1 | 33.0 | 23.9 | 32.2 | 23.6 | 26.0                | 31.8 | 25.3 | 30.9 | 24.8 | 29.9 | 2.5                  | 250  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 24.1                            | 19.2 | 28.2 | 23.8 | 18.8 | 27.8 | 23.0 | 17.9 | 26.7 | 80.5          | 31.8 | 77.4 | 31.1 | 75.3 | 29.7 | 28.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |         |         | Extreme Annual Temperature |          |                    |         | n-Year Return Period Values of Extreme Temperature |          |            |          |            |          |            |          |    |    |
|-------------------|---------|---------|----------------------------|----------|--------------------|---------|----------------------------------------------------|----------|------------|----------|------------|----------|------------|----------|----|----|
|                   |         |         | Mean                       |          | Standard Deviation |         | n=5 years                                          |          | n=10 years |          | n=20 years |          | n=50 years |          |    |    |
| 1%                | 2.5%    | 5%      | Min                        | Max      | Min                | Max     | Min                                                | Max      | Min        | Max      | Min        | Max      | Min        | Max      | DB | WB |
| (a) 7.6           | (b) 6.4 | (c) 5.4 | (e) -9.5                   | (f) 36.3 | (g) 1.8            | (h) 2.2 | (i) -10.9                                          | (j) 37.9 | (k) -11.9  | (l) 39.2 | (m) -12.9  | (n) 40.4 | (o) -14.3  | (p) 42.0 |    |    |
| (4) 7.6           | (5) 6.4 | (5) 5.4 | (4) -9.5                   | (4) 36.3 | (4) 1.8            | (4) 2.2 | (4) -10.9                                          | (4) 37.9 | (4) -11.9  | (4) 39.2 | (4) -12.9  | (4) 40.4 | (4) -14.3  | (4) 42.0 |    |    |

## Monthly Climatic Design Conditions

|      |                                                                   | Annual      |       |       |       |       |       |       |       |       |       |       |       |       |      |
|------|-------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|      |                                                                   | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   |       |      |
| (6)  | Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 15.9  | 5.2   | 6.4   | 10.7  | 15.6  | 19.8  | 24.3  | 25.9  | 25.8  | 22.2  | 16.3  | 11.5  | 6.6  |
| (7)  |                                                                   | DBStd       | 8.57  | 6.17  | 4.66  | 5.05  | 4.64  | 3.79  | 2.98  | 2.45  | 2.58  | 3.13  | 4.56  | 4.85  | 4.98 |
| (8)  |                                                                   | HDD10.0     | 530   | 177   | 119   | 54    | 6     | 0     | 0     | 0     | 0     | 4     | 41    | 129   |      |
| (9)  |                                                                   | HDD18.3     | 1797  | 407   | 335   | 242   | 109   | 31    | 1     | 0     | 7     | 96    | 207   | 363   |      |
| (10) |                                                                   | CDD10.0     | 2686  | 29    | 18    | 74    | 173   | 303   | 428   | 493   | 489   | 367   | 198   | 87    | 24   |
| (11) |                                                                   | CDD18.3     | 911   | 1     | 1     | 5     | 26    | 76    | 179   | 235   | 231   | 124   | 31    | 3     | 1    |
| (12) |                                                                   | CDH23.3     | 8076  | 2     | 2     | 53    | 271   | 661   | 1638  | 2217  | 2121  | 875   | 209   | 26    | 3    |
| (13) |                                                                   | CDH26.7     | 3129  | 0     | 0     | 5     | 73    | 189   | 667   | 952   | 922   | 271   | 49    | 1     | 0    |
| (14) | Wind                                                              | WSAvg       | 2.2   | 2.6   | 2.6   | 2.7   | 2.7   | 2.3   | 2.0   | 1.8   | 1.7   | 2.0   | 2.0   | 2.1   | 2.4  |
| (15) | Precipitation                                                     | PrecAvg     | 1101  | 84    | 64    | 107   | 77    | 84    | 104   | 122   | 111   | 111   | 83    | 77    | 83   |
| (16) |                                                                   | PrecMax     | 1434  | 156   | 137   | 206   | 188   | 150   | 261   | 206   | 273   | 479   | 172   | 206   | 154  |
| (17) |                                                                   | PrecMin     | 868   | 29    | 13    | 23    | 30    | 22    | 39    | 52    | 16    | 3     | 0     | 14    | 22   |
| (18) |                                                                   | PrecStd     | 160   | 34    | 30    | 43    | 36    | 37    | 62    | 41    | 58    | 97    | 43    | 47    | 32   |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4% DB     | 22.5  | 22.8  | 27.4  | 31.4  | 32.3  | 36.1  | 35.6  | 37.0  | 33.0  | 30.9  | 26.1  | 22.8  |      |
| (20) |                                                                   | 0.4% MCWB   | 16.5  | 15.1  | 17.3  | 19.5  | 22.2  | 23.1  | 25.1  | 25.1  | 22.5  | 22.7  | 18.4  | 16.2  |      |
| (21) |                                                                   | 2% DB       | 20.5  | 19.8  | 24.9  | 28.7  | 30.7  | 33.6  | 34.0  | 34.7  | 31.4  | 27.7  | 23.8  | 20.0  |      |
| (22) |                                                                   | 2% MCWB     | 15.8  | 13.3  | 16.3  | 17.6  | 21.2  | 23.1  | 24.7  | 24.6  | 21.7  | 20.3  | 17.3  | 16.4  |      |
| (23) |                                                                   | 5% DB       | 17.8  | 17.6  | 22.3  | 27.0  | 28.7  | 32.2  | 32.8  | 32.9  | 29.4  | 25.9  | 22.2  | 17.4  |      |
| (24) |                                                                   | 5% MCWB     | 14.2  | 12.2  | 14.8  | 17.7  | 20.6  | 23.0  | 24.6  | 24.2  | 21.5  | 19.0  | 16.7  | 14.3  |      |
| (25) |                                                                   | 10% DB      | 15.1  | 15.3  | 19.7  | 24.0  | 27.3  | 30.8  | 31.8  | 31.6  | 27.8  | 23.4  | 19.8  | 15.1  |      |
| (26) |                                                                   | 10% MCWB    | 11.6  | 9.9   | 13.8  | 16.4  | 19.7  | 22.6  | 24.1  | 24.0  | 21.5  | 18.7  | 15.9  | 11.3  |      |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4% WB     | 18.3  | 16.9  | 18.8  | 20.8  | 23.8  | 25.5  | 27.3  | 27.2  | 25.2  | 23.5  | 21.0  | 18.6  |      |
| (28) |                                                                   | 0.4% MCDB   | 20.8  | 19.9  | 23.9  | 28.0  | 29.9  | 31.5  | 33.2  | 32.8  | 29.0  | 27.3  | 22.7  | 20.8  |      |
| (29) |                                                                   | 2% WB       | 16.4  | 15.3  | 17.5  | 19.8  | 22.4  | 24.8  | 26.1  | 26.1  | 24.2  | 22.1  | 19.1  | 17.0  |      |
| (30) |                                                                   | 2% MCDB     | 19.5  | 18.3  | 21.6  | 26.6  | 28.2  | 30.4  | 31.8  | 32.2  | 27.9  | 25.5  | 21.8  | 18.9  |      |
| (31) |                                                                   | 5% WB       | 14.8  | 13.2  | 16.5  | 18.6  | 21.4  | 24.2  | 25.4  | 25.3  | 23.5  | 21.0  | 17.8  | 15.0  |      |
| (32) |                                                                   | 5% MCDB     | 17.8  | 17.2  | 21.3  | 24.0  | 26.9  | 29.4  | 31.0  | 31.1  | 26.8  | 24.2  | 20.3  | 17.5  |      |
| (33) |                                                                   | 10% WB      | 12.3  | 10.4  | 14.9  | 17.6  | 20.5  | 23.6  | 24.8  | 24.7  | 22.7  | 19.6  | 16.3  | 12.2  |      |
| (34) |                                                                   | 10% MCDB    | 14.4  | 14.6  | 19.1  | 22.0  | 25.5  | 28.4  | 30.1  | 29.8  | 25.8  | 22.6  | 19.6  | 14.5  |      |
| (35) | Mean Daily Temperature Range                                      | MDBR        | 9.9   | 11.0  | 11.8  | 12.0  | 11.4  | 10.4  | 9.9   | 10.1  | 10.0  | 10.9  | 11.3  | 10.3  |      |
| (36) |                                                                   | 5% DB MCDBR | 12.4  | 14.5  | 14.6  | 14.8  | 13.0  | 12.3  | 11.2  | 12.0  | 11.8  | 12.6  | 13.3  | 13.3  |      |
| (37) |                                                                   | 5% DB MCWBR | 8.9   | 9.8   | 7.9   | 6.6   | 5.8   | 4.4   | 4.2   | 4.2   | 4.7   | 6.0   | 7.9   | 10.2  |      |
| (38) |                                                                   | 5% WB MCDBR | 11.7  | 12.6  | 12.9  | 12.7  | 11.7  | 10.0  | 10.1  | 10.3  | 8.9   | 9.6   | 10.7  | 11.8  |      |
| (39) |                                                                   | 5% WB MCWBR | 9.0   | 10.3  | 9.4   | 6.8   | 5.9   | 4.3   | 4.3   | 4.3   | 4.6   | 5.4   | 7.4   | 10.5  |      |
| (40) | Clear-Sky Solar Irradiance                                        | taub        | 0.302 | 0.317 | 0.347 | 0.391 | 0.438 | 0.487 | 0.515 | 0.514 | 0.404 | 0.358 | 0.320 | 0.304 |      |
| (41) |                                                                   | taud        | 2.527 | 2.485 | 2.433 | 2.315 | 2.210 | 2.122 | 2.063 | 2.052 | 2.356 | 2.471 | 2.532 | 2.560 |      |
| (42) |                                                                   | Ebn at Noon | 899   | 924   | 920   | 891   | 850   | 806   | 780   | 771   | 850   | 864   | 869   | 874   |      |
| (43) |                                                                   | Edh at Noon | 79    | 93    | 107   | 127   | 143   | 156   | 165   | 163   | 114   | 93    | 78    | 72    |      |
| (44) | All-Sky Solar Radiation                                           | RadAvg      | 2.47  | 3.18  | 4.17  | 5.30  | 5.91  | 6.38  | 6.03  | 5.46  | 4.47  | 3.58  | 2.73  | 2.17  |      |
| (45) |                                                                   | RadStd      | 0.19  | 0.37  | 0.32  | 0.41  | 0.46  | 0.41  | 0.30  | 0.30  | 0.39  | 0.45  | 0.22  | 0.18  |      |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (66 neighbors) | +0.44 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +155    | +106    |

Nomenclature: See separate page